

Geographic Information System

for the Myanmar Operation (GISMO)

GISMO directly links demographic, geographic and project-related information to computerised maps of the project area, providing a unique tool for analysis, assessment, monitoring and reporting.

Technical Overview

The core of GISMO is a MapInfo-based GIS system comprising two levels of administrative boundaries (township and village tract) over base geographic data comprising scanned raster maps, rivers, roads, elevation ranges and village locations.

Administrative boundaries and villages are referenced through a 6 digit coding system and a standardised system for rendering Burmese village and township names into English, allowing information collected by UNHCR and its partners to be imported directly into GISMO and analysed geographically.

GISMO and its related database, ICLPS, were conceived as an integrated tool for managing the Reintegration component of UNHCR's programme to repatriate refugees from camps in Bangladesh to the Rakhine state of Myanmar, since UNHCR's unusual role as the major implementor of assistance projects in the area as well as coordinating agency for the repatriation called for management tools that were not available within the organisation.

Establishing the System

The first step in the process was to establish official administrative boundaries for areas with returning refugees to prevent confusion and ensure that all villages were known to, and covered by, UNHCR field staff. Available data was collected from a number of government agencies and consolidated to form an administrative map of Maungdaw, Buthidaung and Rathedaung townships (which together account for more than 99% of returnees). A standardised system for rendering Myanmar village names into English was developed and agreed upon by UNHCR and its partners along with a coding system that provided each village and administrative area with a unique means of reference.

The completion of this process provided an administrative map covering all villages to which refugees were returning. Demographic information had been collected in parallel and was linked to each area, providing information on local and returnee populations as well as varied data on education, health, agriculture and other sectors. The ability to display and analyse this data geographically was invaluable for planning area-based priorities for protection as well as reintegration activities, and assigning areas of responsibility that took into account such factors as travel impediments, area and concentrations of returnees.

As the administrative data was being collected, a small GIS unit was established in Yangon, staffed by members of the Immigration and Population Department (IMPD) - UNHCR's government counterpart and main implementing partner. Training was provided in the use of ARC/Info software and a 1:64,000 scale map of the project area obtained for digitising. The GIS staff made four coverages, comprising rivers, roads, elevation boundaries and villages.

Once finalised and passed to Maungdaw, this geographic data was ground-checked and calibrated using Global Positioning System (GPS) units - hand-held devices that use satellites to provide precise location information. The administrative boundary data, which at this point is not geographically referenced, will then be overlaid onto the spatial data digitised in Yangon and given geographic co-ordinates. This process will be the final step in combining roads, rivers, and villages with their administrative boundaries and will allow more complex analysis than is possible with either set of data separately.

Integrated Country Level Programme System (ICLPS)

ICLPS centralises project-related information, automates and organises many of the steps involved in project implementation and provides a powerful monitoring and reporting tool.

Overview

As GISMO was being developed, an additional database was established to provide the complex management tools needed to implement the Reintegration programme. With UNHCR acting as both the coordinating agency and main implementor of reintegration activities, a database was needed that could manipulate thousands of activities implemented by hundreds of contractors throughout the project area, at a level of detail that was beyond the capability of the UNHCR budgeting software - FMIS. Thus ICLPS (Integrated Country-Level Programme System) was developed, designed in such a way that all project-related data was linked through the coding system developed for GISMO, and thus capable of being sorted, displayed and analysed geographically.

ICLPS is used at every stage of a project's life, from proposal, approval and contract preparation to instalment payment, monitoring and reporting. Information can be easily sorted and analysed using any combination of fields, producing reports and queries of considerable flexibility and diversity. Examples of queries that can be performed using ICLPS and GISMO include:

- * *Which activities are implemented by a certain contractor for which all instalments have been paid but work is not yet complete?*
- * *Which health activities are not completed one month or more after the contract end date?*
- * *How many children of primary-school age live within the village of a proposed school site?*
- * *How many villagers live within 10kms of a proposed health centre?*
- * *Which contractors have been most successful in carrying out activities within the agreed time frame and budget?*
- * *In which areas has there been a high incidence of wells with brackish water, and what is their average water level?*

Reporting information is entered directly into the system by Field Staff using a simple peer-to-peer Local Area Network and is used by the Programme Unit to analyse progress of projects geographically, sectorally and individually. Information on instalments paid to contractors and village development committees is imported from IMPD into ICLPS weekly, but will eventually be entered directly with expansion of the network. Regular importing and exporting of data between the offices in Maungdaw and Buthidaung ensure both offices have access to the latest information.

All financial information is linked to FMIS sector-activity codes, allowing a detailed analysis of financial obligations during the preparation of submissions and reports.

Components of ICLPS

Proposals: Field Officers or Consultants wishing to propose projects for implementation submit them to the Programme Unit on a defined form, giving a brief project description along with the project location comprising township, village tract, hamlet and location within the hamlet (school, health facility etc.). This data is entered directly into ICLPS giving immediate geographic reference to proposals, and starting a system of tracking that traces the status of a project throughout its life. Proposals are checked and the proposal form returned to the submitting officer with individual reference numbers for each proposal.

Approval: Every week a meeting of the Joint Project Assessment Committee (JPAC) is held, comprising staff from UNHCR, IMPD and NASAKA, at which a list of new proposals produced from ICLPS is discussed and approved. Approval dates are entered into ICLPS and copies of the signed list forwarded to field staff for the next stage, contract preparation.

Batch Concept: Due to the large number of projects implemented directly by UNHCR/IMPD (one thousand at the end of February 1996), a system of 'batch' projects was conceived and implemented. Under this system, where a number of easily replicable projects are needed - such as school buildings, wells and latrines - a single project proposal is prepared, giving justification and full details of the project, and submitted for approval. The implementation aspects of the project are then prepared by the appropriate technical/consultant unit and a standardised contract prepared. This contract is then placed as a template on designated computers and, as individual project sites are approved, the field officer responsible for the project prepares a contract for signature. This decentralisation of contract preparation has avoided bottlenecks at the preparation stage, eased the workload on technical and programme staff and ensured that those with the most intimate knowledge of an individual project are the ones preparing the documentation.

Contract Preparation: Standardised templates comprise a first page containing basic project information (parties, dates, amounts, instalments, applicable clauses), a second listing statutory clauses, a third section with project-specific clauses and budget, and a fourth where necessary with technical designs. As far as possible, this information is already on the template, including designs entered through scanning, and the preparing officer need only enter data specific to the project. One *Contract* can comprise more than one *Activity* - for the purposes of this operation defined as a single project at a single site - and both types of data are entered into ICLPS. Storing data at Contract level only would not have provided enough information to ensure effective monitoring and reporting, since one contract could comprise, for example, multiple wells in different locations, or a school construction and training component. ICLPS thus stores each well as a separate Activity, and the school and training component as separate Activities, so that each can have specific monitoring information attached, be sorted according to sector and project type and include accurate location data.

Monitoring: With so many projects being implemented of varying sizes, using standard UNHCR reporting formats would be too cumbersome. Instead, prior to a field visit, staff print a Field Visit Report from ICLPS, which, at village tract level, gives information on demographics, local authorities, NGO activity, protection issues and sectoral issues as well as listing all completed and uncompleted projects. A map of the area can also be printed from GISMO for reference. Following the field visit, monitoring information is entered directly into ICLPS and is used by field staff and the programme unit to track progress.

Reporting: Narrative information entered by field staff following monitoring trips, combined with the powerful analytical features of ICLPS and GISMO provide a wealth of information on the project. Pre-written reports provide general reporting data, including analysis according to FMIS codes, while the software's flexible reporting features allow custom reports to be prepared and printed as required. All information can be analysed and combined with geo- and demographic data under GISMO.

Other Applications

Protection Survey: GISMO was used as the basis for a questionnaire carried out by Field Staff which sought to establish the views of returnees on conditions in Myanmar and, through this exercise, to identify issues of concern to UNHCR and appropriate responses to them. Linking this data to GISMO allowed comparisons to be made of the situation from area to area to show, for example, whether villages under the administration of a particular commander were more prone to protection problems than others. GISMO provided a means of displaying data and ascertaining which areas might require additional attention to protection issues.

Well Survey: GISMO and ICLPS were used to analyse data from a survey undertaken on UNHCR's project to construct open village wells. Some 200 wells were surveyed and the results displayed geographically. The exercise can show areas more susceptible to salt-water intrusion than others (and in which ponds might be more appropriate), average water levels at points throughout the project area, locations that suffered from turbid water and villages in which build quality was poor (which often reflects the availability of poor-quality local materials as well as inexperienced contractors). Based on this data, the project was modified to take into account areas with high salinity, and structural problems on existing wells addressed.

Road Survey: Travel in the project area is often very arduous. Only one stretch of sealed road exists (linking Maungdaw and Buthidaung towns), with other roads mostly unusable during the rainy season due to damaged and missing bridges and sections of road prone to flooding. To increase access for UNHCR and its partners to villages with high returnee concentrations and improve inter-village access for the local population, UNHCR and WFP worked with local government agencies to survey existing roads and identify critical sections for repair or construction. GPS units were used to map routes and individual bridges, culverts and fords of the most important roads in the area. Data collected included the type, condition and length of river crossings along with an estimated budget for their repair or construction. This information is used to plan and implement maintenance of critical structures to ensure year-round access. Bridges are assigned to different agencies for repair or construction according to their importance and progress mapped on GISMO, allowing a visual representation of the road to be displayed showing bridges which are unpassable, under repair/construction or in good order. GISMO and GPS are essential components in planning an integrated and effective approach to upgrading this and other roads in the area.

Summary

UNHCR's mandate requires it to respond to emergency situations with fast and effective assistance, organise elaborate repatriations within limited timeframes and implement complex reintegration programmes requiring a high degree of flexibility and technical aptitude. The only component most of UNHCR's operations have in common is the need for high quality, versatile, effective resources - from staff to vehicles and from radios to computers.

The dramatic improvement of information technology in recent years has brought previously complex and expensive applications such as GIS, satellite positioning, advanced databases and networking into the reach - both in terms of cost and ease of use - of even small organisations and offices. These applications can simplify complex tasks and provide powerful tools for improving coordination between agencies, site planning, camp management, registration, operational planning, logistics, monitoring, evaluation and reporting. GIS technology especially provides a potential that UNHCR and other field-oriented organisations, dealing in facts of geography and demography every hour of every day, cannot afford to ignore.

Although still a relatively new technology, it is hoped that GIS and other modern tools will quickly become an integral component of UNHCR's operational methodology.

